INCH-POUND
MIL-DTL-24643/31G
27 November 2023
SUPERSEDING
MIL-DTL-24643/31F
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 600 VOLTS, TYPE LS2SU (INCLUDING VARIATION LS2SUS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in two variations: LS2SU and LS2SUS (double overall shield).

Construction (Non-Watertight)

First	-	Copper conductor, coated or uncoated, ASTM B286, 22 AWG with 7 strands.
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation. Nominal wall thickness of 0.013 inch; wall thickness may vary from nominal as necessary in order for the completed cable to meet the specified electrical requirements. Colored insulation, one white and one black conductor for each pair.
Fourth	-	Two conductors (one black and one white) cabled with a nominal lay of 2.5 inches to form a pair.
Fifth	-	An optional binder.
Sixth	-	Braided shield of 34 or 36 AWG, tin-coated copper. Braid angle of 20 to 40 degrees; minimum coverage of 85 percent.
Seventh	-	Shield insulation of two polyester tapes, outer tape sealed. An alternate shield insulation may consist of one polyester tape, plus an extruded jacket of a suitable transparent material; minimum average thickness of the transparent material 0.003 inch. Standard identification code shall be applied by method 2.
Eighth	-	The specified number of pairs (see table I or table II) shall be cabled together in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers may be used as necessary to form a firm, well-rounded assembly.
Ninth	-	An optional binder.

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Construction for Variant Type LS2SU

Tenth - Cross-linked polyolefin jacket. See table I for thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Construction for Variant Type LS2SUS

Tenth - Braided shield of tin-coated copper.

Eleventh - Spirally wrapped, 0.002-inch-thick polyethylene terephthalate (polyester film) separator tape with 25 percent minimum overlap.

Twelfth - Braided shield of tin-coated copper.

Thirteenth - An optional binder.

Fourteenth - Cross-linked polyolefin jacket. See table II for thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

TABLE I. Details (unshielded).

Military part no. M24643/31	Type and size	Number of pairs	Cable jacket thickness min. avg. (inch)	Overall diameter (inches)	
				min.	max.
-01UN	LS2SU-3	3	0.050	0.480	0.520
-02UN	LS2SU-7	7	0.050	0.610	0.660
-03UN	LS2SU-10	10	0.050	0.770	0.830
-04UN	LS2SU-14	14	0.063	0.860	0.930
-05UN	LS2SU-19	19	0.063	0.970	1.040
-06UN	LS2SU-24	24	0.063	1.120	1.210
-07UN	LS2SU-30	30	0.063	1.190	1.280
-08UN	LS2SU-37	37	0.063	1.290	1.380
-09UN	LS2SU-44	44	0.063	1.460	1.550
-10UN	LS2SU-61	61	0.075	1.660	1.740

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TABLE II. Details (shielded).

Military part no. M24643/31	Type and size	Number of pairs	Cable jacket thickness min. avg. (inch)	Overall diameter (inches)	
				min.	max.
-01UD	LS2SUS-3	3	0.050	0.540	0.580
-02UD	LS2SUS-7	7	0.050	0.670	0.720
-03UD	LS2SUS-10	10	0.050	0.830	0.890
-04UD	LS2SUS-14	14	0.063	0.920	0.990
-05UD	LS2SUS-19	19	0.063	1.030	1.100
-06UD	LS2SUS-24	24	0.063	1.180	1.270
-07UD	LS2SUS-30	30	0.063	1.250	1.340
-08UD	LS2SUS-37	37	0.063	1.350	1.440
-09UD	LS2SUS-44	44	0.063	1.520	1.610
-10UD	LS2SUS-61	61	0.075	1.720	1.800

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	17.37
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2,000
Conductor to shield	1,000
Component shield to all other shields	500
Overall shield to overall shield (type LS2SUS only)	200
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	500
Conductor to shield	500
Component shield to all other shields	100
Overall shield to overall shield (type LS2SUS only)	100
Conductor continuity	No failure
Shield continuity	No failure
Jacket flaws (type LS2SUS only)	No failure

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Group A:	
Visual and dimensional	No failure
Capacitance	
Mutual capacitance (nominal at 1 kilohertz [kHz]) (picofarad [pF]/100 feet)	30
Capacitance unbalance (percent, max.)	8
Characteristic impedance, at 1 megahertz (MHz) (ohms)	75±5
Group B:	
Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C]	50
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Attenuation, at 3 MHz (decibel [dB]/100 ft max.)	3.0
Group C:	
Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure

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Shield (conformance to material, construction, and coverage)	No failure
Surface transfer impedance (type LS2SUS only)	
Milliohms per meter (max.)	700
Electromagnetic pulse (EMP) response (dB, min.)	60

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5

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Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Types	Sizes	Feet (nominal)
LS2SU and LS2SUS	3 through 30	1,000
LS2SU and LS2SUS	37 through 61	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-047)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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